

4.2.2 GenAI-Adopting Firms

We define a firm as a GenAI adopter if it has posted at least one GenAI integrator vacancy. By this criterion, 10,433 firms qualify as adopters. While they make up only 3.71 percent of the 281,111 firms in our sample, adopters are disproportionately large (see descriptive statistics below) and account for 16 percent of the employment (positions) in our dataset.

Figure 2 plots the timing of GenAI adoption, defined as the posting date of each firm's first GenAI integrator vacancy. Prior to 2023, adoption was minimal and stable, with roughly 30 new adopters per month. Beginning in early 2023—shortly after the launch of ChatGPT—the number of new adopters rose sharply, peaking at 456 in August 2023. Adoption then stabilized at around 400 firms per month through the end of 2024 before accelerating again in early 2025, reaching 574 new firms in March. By the end of the sample period, the cumulative number of adopters had surpassed 10,000.

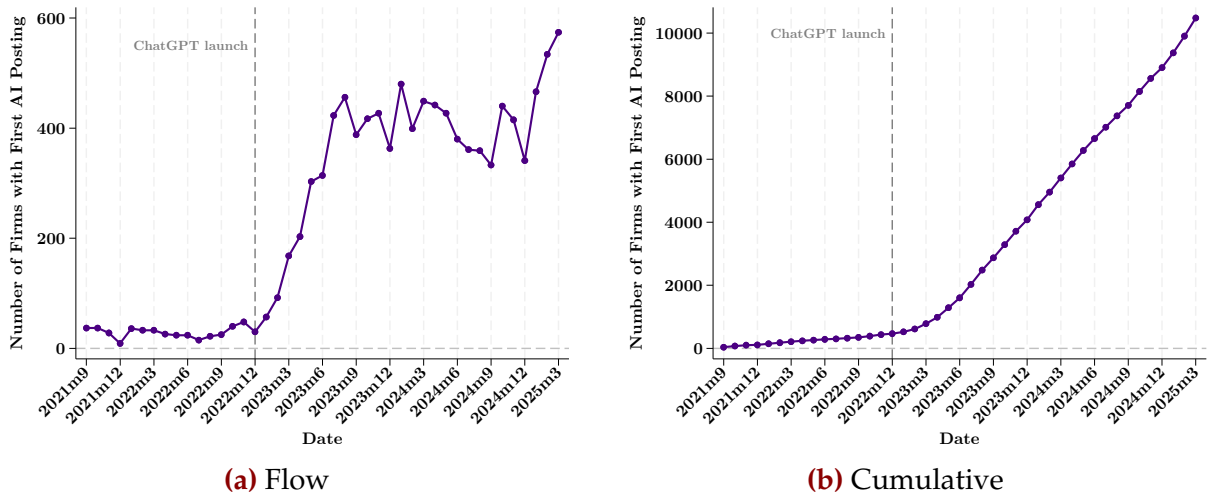


Figure 2: Number of Adopters Over Time

Notes: Panel (a) shows the monthly number of firms posting their first GenAI integrator vacancy, while Panel (b) reports the cumulative total, covering September 2021 to March 2025.

Table A.5, reported in Supplemental Appendix A.15, presents descriptive statistics for the full sample, adopters, and non-adopters. Adopters are substantially larger, more senior-heavy, and recruit more heavily from highly exposed occupations than non-adopters; they are also concentrated in professional services and information and in technology-intensive states, while non-adopters are more widely dispersed across sectors. Supplemental Appendix A.16 provides further detail on the sectoral and geographic distribution

of adopters versus non-adopters.

4.2.3 Validation: GenAI Mentions in Worker Position Descriptions

As external validation of the adoption measure, we examine whether workers at adopting firms are more likely to report AI-related activities in their own LinkedIn position descriptions—an outcome that reflects actual on-the-job use rather than aspirational hiring language.²⁴ Using a list of GenAI keywords (the full list is reported in Supplemental Appendix A.8) applied to 18.3 million position descriptions (2021–2025), we compute for each quarter the share of descriptions mentioning GenAI keywords, separately for adopting and non-adopting firms.

Figure 3 plots these shares, normalized to zero in 2021 Q1. Prior to the launch of ChatGPT in late 2022, both series fluctuate near zero with no visible divergence. Beginning in early 2023, the adopter series rises sharply—reaching approximately 3.5 percentage points above baseline by late 2024—while the non-adopter series increases only modestly (roughly 0.7 percentage points). A difference-in-differences regression with firm and month fixed effects formalizes this pattern and yields an estimate of 1.35 percentage points (s.e. = 0.05, $p < 0.001$), and the coefficient remains essentially unchanged (1.40 percentage points) when additionally controlling for description length (full regression results are reported in Supplemental Appendix A.8). Relative to the pre-ChatGPT average of 3.1% at adopters, this represents a roughly 43 percent increase. This worker-side evidence supports the interpretation that our firm-level adoption classification captures genuine changes in workplace technology use.

4.3 Data Validation

We subject our data and key variables to a series of validation exercises, reported in full in the Supplemental Appendix.

²⁴Position descriptions on LinkedIn are considerably less reliable than job posting texts: only about 42% of U.S. positions in our 2021–2025 window have any raw description text at all, and those that exist are often short or formulaic. For this reason, we rely on the richer and more complete raw posting data to construct the firm-level adoption measure throughout the paper. The position-description analysis presented here serves as a complementary validation exercise; in Supplemental Appendix A.4 we additionally use these descriptions to construct an augmented adoption measure and show that our main results are robust.

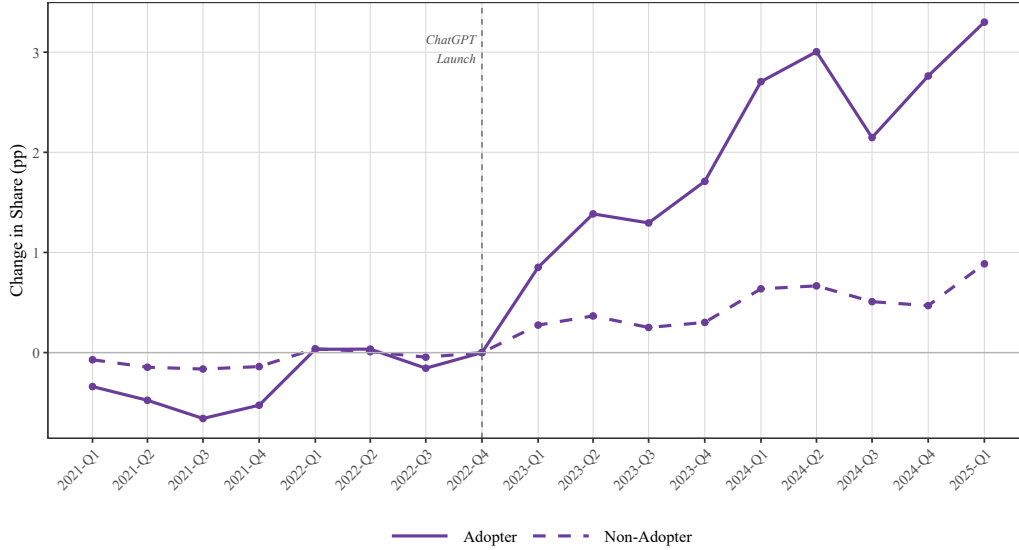


Figure 3: Change in Share of Position Descriptions Mentioning GenAI Keywords

Notes: This figure plots the share of LinkedIn position descriptions containing at least one GenAI keyword, by quarter, separately for GenAI-adopting and non-adopting firms. The sample includes 18.3 million U.S. positions from 2021 Q1 through 2025 Q1 held at firms in the main analysis sample, restricted to positions with a raw description of at least 20 characters. Both series are normalized to zero in 2021 Q1. The vertical dashed line marks the launch of ChatGPT in November 2022. Baseline levels (2021 Q1): Adopters = 2.9%, Non-Adopters = 0.8%.

First, we assess the *representativeness* of the LinkedIn/Revelio data by comparing its industry, occupation, and age distributions against official employment statistics from the QCEW, OEWS, and QWI. The data display the expected tilt of LinkedIn toward managerial, professional, and high-information occupations and industries. However, importantly, this tilt is essentially unchanged from 2021Q1 through 2025Q1, with no discernible break around the ChatGPT launch. Because the level differences are absorbed by firm and sector fixed effects in our main specifications, the differential identification of treatment effects is unaffected (see Appendix A.1 for details).

Second, we verify the *stability of the seniority classification* through two complementary exercises (Appendix A.2). We first analyze the frequency of job-title keywords (e.g., “assistant,” “manager,” “director”) separately for junior and senior workers, and confirm that the classification aligns well with conventional titling. Both pooled pre/post-2023 comparisons and a formal event study at adopting versus non-adopting firms show no meaningful differential shift in keyword distributions around the ChatGPT launch. We

then examine the experience composition of junior and senior workers over time, separately at adopting and non-adopting firms. We find that the average labor market experience of junior and senior workers evolves in parallel at adopting and non-adopting firms; a statistically detectable but quantitatively small differential emerges for seniors in the event-study specification, but its magnitude is unlikely to meaningfully affect our employment results. Together, these exercises indicate that the relative decline in junior employment we find below is unlikely to be driven by changes in seniority classification.

5 Results

The main empirical exercise of the paper is to test the framework’s central implication: whether GenAI adoption disproportionately reduces junior employment relative to senior employment. We pursue this in Sections 5.2–5.5 using the firm-level résumé and posting data, with a triple-difference design, a staggered event study, and a flow decomposition. Before turning to this aggregate evidence, Section 5.1 brings direct empirical evidence on the framework’s underlying channels.

5.1 Empirical Evidence on the Framework’s Channels

Proposition 1 decomposes the change in junior labor demand into three forces: a *displacement* effect ($\alpha > 0$, junior tasks lost to automation), a within-task *productivity* effect (on the workers-per-task margin), and a *reallocation* effect ($\hat{x}' > \hat{x}$, juniors expanding into formerly-senior tasks because $\delta_J > \delta_S$). For the productivity channel, and especially for the assumption $\delta_J \geq \delta_S$, a now-large experimental literature provides strong evidence (Brynjolfsson et al., 2025b; Noy and Zhang, 2023; Dell’Acqua et al., 2023; Cui et al., 2025; Cruces et al., 2026). This subsection tests the remaining two channels using evidence from two sources: task-level information extracted from our own postings data, and external task-level data from the Anthropic Economic Index.